

# FNU MOHBAT

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## SUMMARY

Final year PhD candidate in computer science with research experience in multi-modal learning, large language models (LLMs), knowledge graphs (KGs) and graph neural networks (GNNs). Skilled in training and deploying LLMs, vision-language models (VLMs), and retrieval augmented generation (RAG) on multi-GPU systems. *Looking for a machine learning researcher role to leverage and expand my skills.*

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Ph.D. Computer Science</b> , Rensselaer Polytechnic Institute, Troy, NY<br><i>Dissertation:</i> Knowledge Graph Enhanced Large Language Models for Food Understanding.<br><i>Advisor:</i> Mohammed J. Zaki | 01/2021 - 08/2025                      |
| <b>MS Electrical Engineering</b> , Lahore University of Management Sciences (LUMS)<br><i>Thesis:</i> Content Based Image Retrieval Through Deep Learning.                                                     | 08/2016 - 06/2018                      |
| <b>BS Electrical Engineering</b> , COMSATS Institute of Information Technology<br><b>Undergraduate Exchange</b> , Ivy Tech Community College, Fort Wayne, IN                                                  | 09/2009 - 06/2014<br>08/2012 - 05/2013 |

## SKILLS SUMMARY

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|-------------------------------|---------------------------------------------------------------------------|
| <b>Programming Languages:</b> | Python, C/C++, MATLAB, C#, VB .Net                                        |
| <b>AI Frameworks:</b>         | PyTorch, PyTorch-Geometric, Keras, TensorFlow, langchain, LlamaIndex      |
| <b>CV &amp; Robotics:</b>     | OpenCV, ROS, PCL                                                          |
| <b>Platforms:</b>             | IBM Power 9, HPC, Nvidia Jetson, Odroid, Raspberry Pi, Arduino, Intel80xx |

## RESEARCH EXPERIENCE

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| <b>Rensselaer Polytechnic Institute</b><br><i>Graduate Research Assistant, Advisor: Mohammed J. Zaki</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Troy, NY<br>08/2021 - Present             |
| <ul style="list-style-type: none"><li>Conducting research to enhance Large Language Models (LLMs) and Multi-Modal Models (MMMs) by integrating Knowledge Graphs (KGs) through Retrieval-Augmented Generation (RAG), achieving a 30 – 50% improvement in personalized food recipe recommendations.</li><li>Proposed multistage instruction tuning and custom loss for improving language vision models, resulting a 16-point gain in SacreBLEU and a 6-point increase in ROUGE-L scores.</li><li>Improved document understanding, caption generation, and text summarization through LLM (LLaMA, Phi, Mistral) and MMM (LLaVA, miniGPT).</li><li>Facilitated collaboration between IBM Research and Rensselaer Polytechnic Institute, combining academic and industry expertise to foster innovation (2021-2023).</li></ul> |                                           |
| <b>IBM Thomas J. Watson Research Center</b><br><i>Summer Extern, Mentors: Keerthiram Murugesan</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Remote<br>05/2023 - 08/2023               |
| <ul style="list-style-type: none"><li>Investigated the impact of Stable Diffusion (SD) generated images on visual concept manipulation in Multi-Modal Models (MMMs), revealing that the biased visuals exacerbate toxicity and bias in generated text compared to unbiased visual and text-only inputs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |
| <b>IBM Thomas J. Watson Research Center</b><br><i>Summer Extern, Manager: Ashish Verma</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Yorktown Heights, NY<br>05/2022 - 08/2022 |
| <ul style="list-style-type: none"><li>Optimized generalization of document AI models by 15 – 30% on out-of-distribution (OOD) data through novel Transformer-GNN integration, reducing OOD detection error rates and enhancing real-world adaptability.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |
| <b>IBM Thomas J. Watson Research Center</b><br><i>Summer Extern, Mentors: Ashish Verma, Catherine Finegan-Dollak</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remote<br>05/2021 - 08/2021               |
| <ul style="list-style-type: none"><li>Fine tuned transformer models and object detection models for document text understanding tasks including classification and key-value prediction.</li><li>Annotated a proprietary visual documents dataset (10K samples) and fine-tuned and evaluated models such as Bert, LayoutLM, on it.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |
| <b>Lahore University of Management Sciences</b><br><i>Research Associate in National Agriculture Robotics Lab</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Lahore, Pakistan<br>04/2019 - 12/2020     |
| <ul style="list-style-type: none"><li>Led 5-10 researchers in developing deep learning and IoT solutions for water quality assessment via foreign object detection, implemented on Raspberry Pi and NVIDIA Jetson Nano.</li><li>Researched model compression techniques, developing a Teacher-Class network that significantly enhanced accuracy and reduced parameters by 10-30x compared to traditional single-student methods.</li><li>Co-supervised 4 undergraduate and 2 master's students, drafted proposal for research grant proposals and industrial collaborations.</li></ul>                                                                                                                                                                                                                                    |                                           |

- Proposed a residual dyad network that amplified image retrieval by 22 mAP points on remote sensing images.
- Developed fixed-wing V-TOL UAV for autonomous target identification, payload delivery (5kg, 10km range), with 60-min flight time.
- Wrote modification codes for GTA-V to collect 100,000 road scene images with car parameters, and trained Convolutional Neural Networks (CNNs) to predict speed and steering angles for autonomous vehicle control within the game environment.
- Conducted tutorials in Computer Vision and Deep Learning classes and mentored senior year projects.

## PUBLICATIONS

10. Knowledge Graph-Enhanced LLM for Food Recommendation through Question Answering, Fnu Mohbat & Mohammed J. Zaki. Towards Knowledgeable Foundation Models at AAAI, 2025. [PDF](#).
9. LLaVA-Chef: A Multi-modal Generative Model for Food Recipes, Fnu Mohbat & Mohammed J. Zaki. Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM), 2024. [PDF](#).
8. Beyond Visual Augmentation: Investigating Bias in Multi-Modal Text Generation, Fnu Mohbat, Vijay Sadashivaiah, Keerthiram Murugesan, Amit Dhurandhar, Ronny Luss & Pin-Yu Chen. Fourth Workshop on Trustworthy Natural Language Processing (TrustNLP), at NAACL 2024. [PDF](#).
7. GVdoc: Graph-based Visual Document Classification, Fnu Mohbat, Mohammed J. Zaki, Catherine Finegan-Dollak & Ashish Verma. Findings of the 61<sup>st</sup> Annual Meeting of the Association for Computational Linguistics, 2023. [PDF](#).
6. Teacher-Class Network: A Neural Network Compression Mechanism, Shaiq Munir Malik, Fnu Mohbat, & Murtaza Taj. The 32<sup>nd</sup> British Machine Vision Conference, 2021. [PDF](#).
5. Trash Detection on Water Channels, Mohbat Tharani, Abdul Wahab & Murtaza Taj. International Conference on Neural Information Processing, 2021. [PDF](#)
4. Cross-view Image Retrieval - Ground to Aerial Image Retrieval Through Deep Learning, Numan Khurshid, Talha Hanif, Mohbat Tharani & Murtaza Taj. International Conference on Neural Information Processing (ICONIP), Sydney, Australia. Dec 12-15, 2019. [Link](#)
3. A Residual-Dyad Encoder Discriminator Network for Remote Sensing Image Matching, Numan Khurshid\*, Mohbat Tharani\*, Murtaza Taj & Faisal Qureshi. IEEE Transaction on Geo-science and Remote Sensing, Nov, 2019. [Link](#)
2. Dimensionality Reduction Using Discriminative Autoencoders for Remote Sensing Image Retrieval, Mohbat, Tooba Mukhtar, Numan Khurshid & Murtaza Taj, The 20<sup>th</sup> International Conference on Image Analysis and Processing (ICAIP), Trento, Italy, September 9-13, 2019. [Link](#)
1. Use of Greendrone UAS System for Maize Crop Monitoring, Ahmad Kamal Nasir & Mohbat Tharani<sup>1</sup>, The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 2017. [PDF](#)

## PROFESSIONAL SERVICES

- Invited talk and poster at ACM International Conference on AI in Finance (ICAIF), Brooklyn, NY. 2024
- Served as a session chair at ACM Conference on Information and Knowledge Management (CIKM) 2024
- Reviewer: Computer Vision and Pattern Recognition (CVPR) 2025
- Reviewer: ACM International Conference on Information and Knowledge Management (CIKM) 2024
- Reviewer: ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2024 - 2025
- Reviewer: European Conference on Computer Vision (ECCV) 2024
- Reviewer: IEEE Winter Conference on Applications of Computer Vision (WACV): 2020 - 2024

## HONORS AND AWARDS

- AAAI-2025 Travel/Volunteer scholarship award (\$700 + Registration) 12/2025
- NAACL 2024 registration award for volunteering virtual sessions 06/2024
- Founding member and Vice President, Computer Science Graduate Council at RPI 2021 - 2022
- Qualified UAV Medical Express Challenges 2018 in Australia, only 13 out of 55 teams qualified 09/2018
- Received 90% financial aid for Masters degree at Lahore University of Management Science, Pakistan 2016-2018
- National ICT R&D merit scholarship for the complete bachelor's degree. (120/7000 applicants). 2009-2014
- NESAs UGRAD Exchange Scholarship (fully funded two semesters) (3/5000 applicants) 2012-2013
- Inspiring Intern certificate from the State Department during NESAs exchange program (3 out of 90) 4/2013

## OTHER EXPERIENCE

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### **Rensselaer Polytechnic Institute**, *Teaching Assistant*

- Computational Vision (CSCI-6270); Instructor: Chuck Stewart 01/2025 - 05/2025
- Computer Science I (CSCI-1100); Instructor: Uzma Mushtaque 08/2024 - 12/2024
- Operating Systems (CSCI-4210); Instructor: David Goldschmidt 05/2024 - 08/2024
- Computer Organization (CSCI-2500); Instructor: Prof. George Slota

### **Lahore University of Management Sciences**, *Teaching Assistant*

- Deep Learning (CS-437/5317); Instructor: Prof. Murtaza Taj Spring 2019, Spring 2020
- Robot Motion Planning (EE-562); Instructor: Prof. Muhammad Abubakr Fall 2018

### **Ivy Tech Community College** *Teaching Assistant*

- Introduction to database: Oracle 11g; Instructor: Dr. John M. Heise 09/2012 - 05/2013
- Introduction Industrial Technology; Instructor: Dorothy R. Barse Fall 2012
- Introduction Industrial Technology; Instructor: Dorothy R. Barse Fall 2012, Spring 2013

### **Agatos Pvt. Pakistan**, *Automation Engineer*

- Analyzed project requirements, programmed PLCs, HMI/SCADA systems and commissioned on site, and participated in sales meetings with customers 08/2014 - 10/2016